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Beyond the Beat: Audio and Lyric Contributions to Music Classification

Beats & Bars

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Abstract

Music streaming platforms use machine learning models for tasks such as genre classification, recommendation, and content moderation. While most systems rely on measurable audio features, musical identity is also shaped by lyrics, language, and cultural context. This project examines the relative and combined contributions of Spotify audio features and song lyrics in predictive modeling. Using a large-scale dataset of approximately 550,000 tracks with audio attributes, lyrics, genre labels, and explicit content indicators, we address two questions. First, we evaluate genre classification performance using audio-only and multimodal models and assess the contribution of each feature source. Second, we investigate whether explicit content can be predicted from audio features alone and whether explicit tracks differ systematically in their acoustic properties. Interpretable models and tree-based ensembles are compared to assess predictive performance. The results highlight the strengths and limitations of audio-only and multimodal approaches in real-world music analytics.

Introduction

Music streaming platforms rely heavily on audio features such as tempo, loudness, and energy for tasks including genre classification and content moderation. However, these features may not fully capture lyrical themes and linguistic context that also shape musical style. Genre classification remains challenging due to overlapping styles and inconsistent labeling, and lyrics may provide complementary information for distinguishing acoustically similar genres. Additionally, while explicit content labels are largely lyric-based, it is unclear whether explicit tracks exhibit systematic differences in audio features. Using a large-scale Spotify dataset with audio features, lyrics, genre labels, and explicit content indicators, this project addresses two tasks. First, we compare audio-only, lyrics-only, and multimodal models for multiclass genre classification. Second, we examine whether explicit content can be predicted from audio features alone and whether explicit and non-explicit tracks differ in their acoustic properties. This study assesses the relative value of audio and lyrical information in music analytics and recommendation systems.

Genre Prediction

We evaluated three supervised multiclass classification models for music genre prediction using the Spotify dataset, an audio-only model, a lyrics-only model, and a multimodal model that combined both audio features and lyrical content. All models were trained and evaluated using a stratified 70/30 train-test split. Performance was assessed on the test set using overall accuracy, precision, recall, and F1 score.

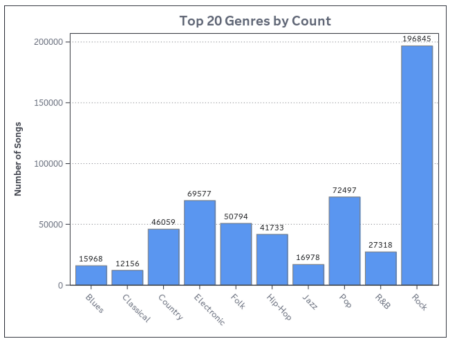

Figure 1. Distribution of songs across the top 20 genres

Figure 1 illustrates the distribution of songs across the top genres in the dataset, revealing a pronounced class imbalance.

Audio-only Model

The audio-only model used a Random Forest classifier trained on standard Spotify acoustic features. This model captured genre-specific acoustic patterns and achieved moderate classification performance. The audio-only model achieved a moderate overall accuracy of 47%, but substantially lower precision, recall, and F1 score, 39%, 27%, and 29% respectively, indicating uneven performance across genres. This behavior is consistent with the strong class imbalance observed in Figure 1. Overall, these findings indicate that while audio features are informative for genre classification, the effectiveness of an audio-only approach is strongly influenced by class-distribution.

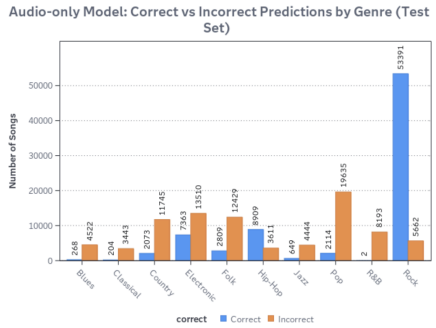

Figure 2. Correct v. Incorrect Predictions by Genre for the Audio-only model

Across most genres, the audio-only model produces a substantial number of correct predictions, particularly for genres with distinctive acoustic profiles such as Electronic, Pop, Hip-Hop, and Rock. Compared to the lyric-only model, correct predictions are more evenly distributed across genres. Overall, this visualization supports quantitative metrics for the audio-only model, moderate accuracy with improved recall across genres, but uneven performance driven by dataset imbalance.

Lyric-only Model

The lyric-only model represented song using hashed bag-of-words features derived from cleaned lyrical text and was trained using a Gradient Boosting classifier. While this model demonstrated the ability to learn genre-specific linguistic cues, its performance was lower than the audio-based approach. This model exhibits a lower overall accuracy, 36%, compared to the audio-only model, indicating that textual information alone is less effective for predicting genre. Although the precision is similarly low, 36%, the most notable

limitation is the extremely low recall, 10%, which suggests that the model correctly identifies only a small fraction of songs belonging to each genre. Despite a relatively higher F1 score, 53%, this value should be interpreted cautiously, as it may be driven by performance on a small number of dominant classes rather than balanced classification across genres.

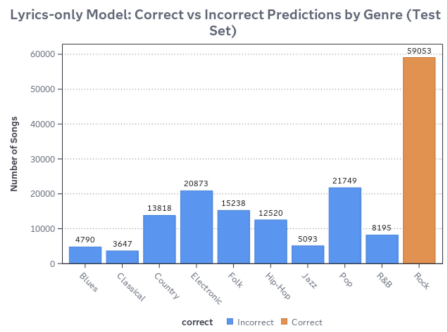

Figure 3. Correct v. Incorrect Predictions by genre for the Lyric-only model

Figure 3 demonstrates that nearly all correct predictions come from the Rock genre, while every other genre is overwhelmingly misclassified. This could be due to Rock being the most frequent genre in the dataset. The metrics and visualization demonstrate that the lyrics-only model is heavily biased by genre imbalance and lacks sufficient discriminative power to separate genres based solely on text features.

Multimodal (Audio + Lyrics) Model

The multimodal model, which integrated both audio features and hashed lyric representation within a single Gradient Boost framework, achieved the highest overall accuracy among the three approaches. These findings support the use of multimodal representations for music classification tasks and demonstrate that incorporating lyrics enhances genre prediction beyond audio features alone. This model demonstrates the strongest overall performance among the three approaches, achieving an accuracy of 54%. Improvements are observed across all metrics relative to the audio-only and lyric-only models, with higher precision, 43%, recall 38%, and F1 score, 39%. The increase in recall suggests that the multimodal model is more effective at identifying a wider range of genres, particularly those that may be underrepresented or acoustically similar.

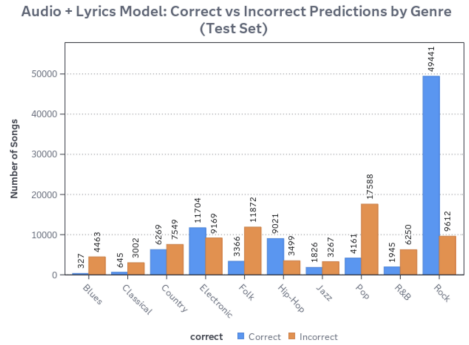

Figure 4. Correct v. Incorrect Predictions by genre for the Multimodal model

Figure 4 demonstrates that the multimodal model shows clear improvements for genres with strong audio and lyrical signals. Rock stands out with the highest number of correct predictions, indicating that combining instrumentation patterns with lyrical content is effective for this genre. Overall, the figure highlights that while multimodal learning

improves performance for some genres, however, class imbalances and genre overlap still limit consistent gains across all categories.

Model Comparison

Model	Accuracy	Presicion	Recall	F1 Score
Audio-only	0.47112	0.38915	0.26700	0.28677
Lyric-only	0.3579	0.3579	0.1000	0.5272
Multimodal	0.5378	0.4345	0.3813	0.3871

Table 1. Model comparison by accuracy, precision, recall, and F1 score.

The audio-only, lyric-only, and multimodal models each show different strengths and limitations. Audio-only models capture general sound characteristics, but have difficulty separating genres that sound similar. Lyric-only models are effective at identifying themes, language, and explicit content, but perform less well for genres defined mainly by musical style rather than words. Multimodal models combine both audio and lyrical information and achieve the best overall performance by using the strengths of each approach. This comparison shows that audio and lyrics provide complementary information, and that combining them leads to more accurate classification.

Explicit Content Prediction

To test whether audio features alone can predict explicit content, we trained two binary classifiers on a 70/30 stratified split using a target flag derived from lyrics, yielding 124,684 non-explicit and 40,293 explicit tracks (approx. 3:1 imbalance). Models used only Spotify audio features and were evaluated at both the default 0.50 threshold and a lowered 0.30 threshold to balance precision and recall for the minority class.

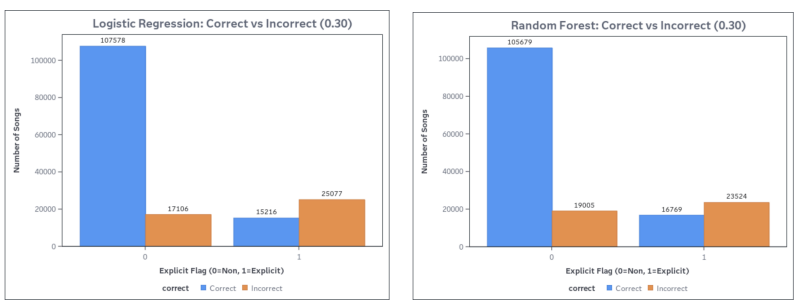

Figure 6. Correct vs. Incorrect Predictions by class for Random Forest and Logistic Regression

Both graphs show the classification performance of Random Forest and Logistic Regression at a lowered 0.30 threshold. In both models, the majority non-explicit class is predicted much more accurately than the minority explicit class, reflecting the dataset's imbalance. Random Forest correctly classifies slightly more explicit tracks and fewer non-explicit tracks compared with Logistic Regression, indicating a modest improvement in recall for the minority class. Logistic Regression, on the other hand, misclassifies more explicit tracks while slightly over-predicting the non-explicit class, highlighting its limitations in capturing nonlinear interactions. Overall, Random Forest demonstrates a more balanced performance,

particularly for the underrepresented explicit tracks, while both models still struggle with minority class prediction.

Feature Importance

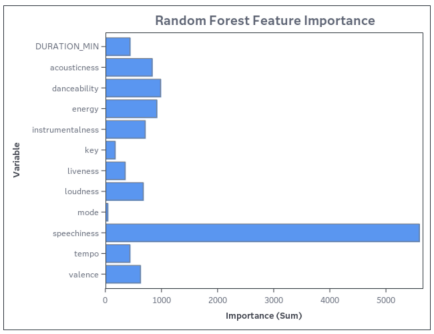

Figure 7. Random Forest Feature Importance

Figure 7 shows that speechiness, Spotify's measure of how much spoken word is present in a track, overwhelmingly dominates the Random Forest model, with an importance score around 5,500, far exceeding all other features. Danceability, acousticness, and energy contribute modestly, while mode and key add little. This indicates that speech-forward production is the primary audio signal distinguishing explicit content, while the remaining audio features provide only limited additional predictive value.

Model Comparison

Model	Threshold	Accuracy	Precision	Recall	F1
Logistic Regression	0.50	0.771	0.633	0/136	0.223
Logistic Regression	0.30	0.745	0.471	0.384	0.423
Random Forest	0.50	0.779	0.673	0.18	0.284
Random Forest	0.30	0.745	0.474	0.414	0.442

Table 2. Performance metrics for Logistic Regression and Random Forest models

Random Forest consistently performs slightly better than Logistic Regression at equivalent thresholds, and lowering the threshold to 0.30 produces more balanced and practically useful predictions than the default 0.50. The class imbalance between non-explicit and explicit tracks suppresses recall at the higher threshold, as both models tend to favor the majority class during training. Overall, audio features provide a limited but meaningful signal for explicit content, largely driven by speechiness, but cannot fully capture explicitness without incorporating direct lyrical analysis.

Conclusion

This study explored using Spotify audio features and lyrics for genre classification and explicit content prediction. The combined audio+lyrics model achieved the highest genre accuracy (54%), outperforming audio-only (47%) and lyrics-only (36%) models, though class imbalance and genre overlap limited performance. For explicit content, audio features provided a modest signal (~77% accuracy), with speechiness as the key predictor, but recall improved only with a lower threshold. Overall, audio captures broad acoustic patterns but is limited for lyric-driven tasks, highlighting the need for richer text features and advanced modeling.

References

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